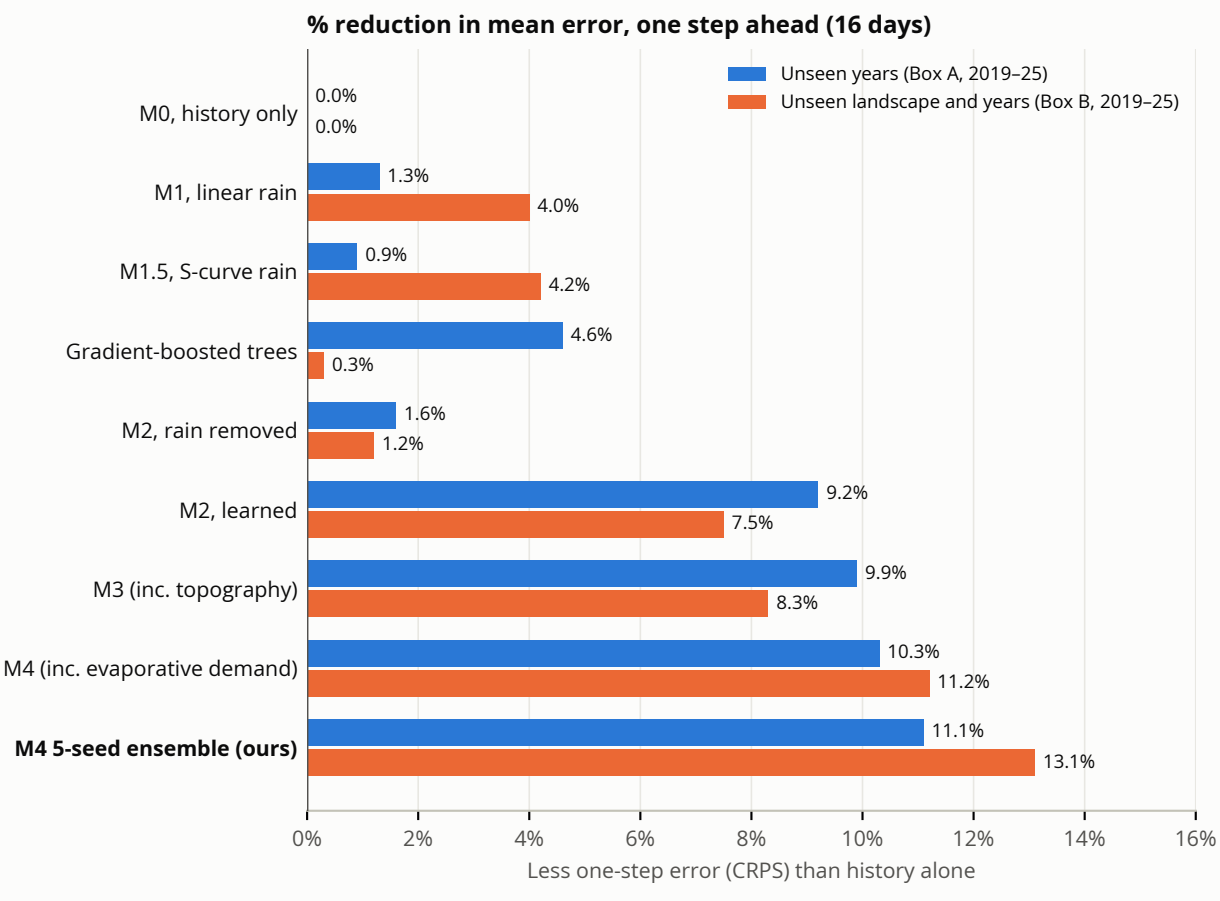
% reduction in mean error, one step ahead (16 days)
Unseen years (Box A, 2019–25)
Unseen landscape and years (Box B, 2019–25)
M0, history only
0.0%
0.0%
M1, linear rain
1.3%
4.0%
M1.5, S-curve rain
0.9%
4.2%
Gradient-boosted trees
4.6%
0.3%
M2, rain removed
1.6%
1.2%
M2, learned
9.2%
7.5%
M3 (inc. topography)
9.9%
8.3%
M4 (inc. evaporative demand)
10.3%
11.2%
M4 5-seed ensemble (ours)
11.1%
13.1%
0%
2%
4%
6%
8%
10%
12%
14%
16%
Less one-step error (CRPS) than history alone